with some basics to understand these advantages and the underlying mechanism. gRPC consists of two components: Protocol Buffers and the classic remote procedure call (RPC).

NOTE The source code for this section is available at www.manning.com/books/akka-in-action-second-edition or https://github.com/franciscolopez sancho/akka-topics/tree/main/chapter11b. You can find the contents of any snippet or listing in the .scala file with the same name as the class, object, or trait.

11.2.1 The Protocol Buffers side

Protocol Buffers is an interface definition language (IDL) that lets you define data structures without being tied to a specific language implementation like Scala, Go, Java, or Python. You define your data structures in a .proto file. The protobuf compiler libraries read this file and create the corresponding data structures in the implementation language of your choice. In Java, for example, these are classes with getters and builders instead of setters.

Serializers/deserializers are generated for you. Protocol Buffers has an efficient format compression rate and high serialization/deserialization performance. Last but not least, its schema development is flexible and robust, which is a crucial aspect for the long-term development of an application.

11.2.2 The RPC side

The RPC part enables HTTP/2 communication with multiplexing. This means you can have data belonging to multiple client/server